Mark schemes

Q1.

(a) $10 = 0 \times 2.5 + \frac{1}{2} a \times 2.5^2$

M1: Use of constant acceleration equation to find a with u = 0.

M1

A1: Correct equation.
NOTE: If v is found first, do not award any marks for part (a) until an equation to find a is produced. This could be from graphical method or from the use of

$s = \frac{1}{2}(u + v)t$.

A1

$a = \frac{20}{2.5^2}$ = 3.2 m s^{-2}

A1: Correct acceleration

A1

3

(b) $10 = \frac{1}{2}(0 + v) \times 2.5$

M1: Use of constant acceleration equation to find v with u = 0.

M1

$v = 8$ m s^{-1}

A1: Correct speed from correct working.

A1

OR

$10 = \frac{1}{2} v \times 2.5$

v = 8 m s^{-1}

NOTE: If v is found in part (a), with correct working award full marks.

OR

$v^2 = (0^2 +)\ 2 \times 3.2 \times 10$

NOTE: Accept $3.2 \times 2.5 = 8$

$v = 8$ m s^{-1}

OR

$v = (0 +)\ 3.2 \times 2.5 = 8$ m s^{-1} **AG**

2

(c) $t = \dfrac{90}{8}$ = 11.25 s

B1: Calculation of correct additional time. Could be implied by later working.

B1

Total Time = 2.5 + 11.25
= 13.75
= 13.8 s

M1: Addition of their time for the 90 metres and the 2.5 seconds.

M1

A1: Correct total time. Accept 13.75.

A1

NOTE: 22.5 + 2.5 = 25 scores B0M1A0

3

(d) $\dfrac{100}{13.75} = 7.27$ m s^{-1}

M1: Finding average speed. Must see 100 and their answer from part (c).

M1

A1F: Follow through candidate's time from part (c), regardless of working in part (c). Allow 7.25 m s^{-1} from 13.8 seconds.

A1F

2

[10]

Q2.

(a) $v^2 = 0^2 + 2 \times 9.8 \times 15$

M1: Use of constant acceleration equation to find v with u = 0 and a = ± 9.8.

M1

$v^2 = 294$

A1: Correct equation

A1

$v = 17.1$ m s^{-1}

A1: Correct speed from correct working. Accept AWRT 17.1. Accept 17.15.

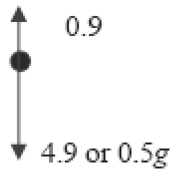
Accept $7\sqrt{6}$

Note: If g = 9.81 is used for the first time deduct one mark. Should get 17.2 m s^{-1} from g = 9.81.

A1

3

(b) (i)



B1: Correct diagram, with arrows and labels. Must see 0.9 and 4.9 or 0.5g (or 4.905 if working with g = 9.81).

B1

1

(ii) $4.9 - 0.9 = 0.5a$

M1: Uses 0.5a.
B1: Explicit statement of "4.9 – 0.9" or "mg – 0.9" or "0.5g – 0.9".

M1B1

$(a =)\dfrac{4}{0.5} = 8$ ms^{-2} **AG**

A1: Correct acceleration from correct working. Can be awarded without the B1 mark.

Must see $\dfrac{4.9(\text{or } 0.5g) - 0.9}{0.5}$ or $\dfrac{4}{0.5}$ *or 4 = 0.5a*

A1

Note: If g = 9.81 is used candidates will get 8.01 m s^{-2}. Deduct 1 mark if 8.01 is seen.

Examples:

$4.9 = 0.5a + 0.9$ M1 B0 A0

$a = 8$

$4 = 0.5a$ *M1 B0 A1*
$a = 8$
If candidates only write

$a = \frac{4}{0.5} = 8$ *award M0 B0 A0.*

3

(iii) $v^2 = 0^2 + 2 \times 8 \times 15$

M1: Use of constant acceleration equation to find v with $u = 0$ and $a = \pm 8$.

M1

$v = 15.5$ m s^{-1}

A1: Correct speed from correct working.
Accept AWRT 15.5 or truncated to 15.4.
Accept $4\sqrt{15}$.

A1

2

(iv) The air resistance force will not be constant, but changes **as the speed of the ball changes** (or **changes as the ball accelerates**).

B1: Correct explanation, key words in bold.
Do not award mark for statements that imply that the acceleration causes the air resistance to change.

B1

1

[10]

Q3.

(a) $(8\mathbf{i} + 12\mathbf{j}) + (4\mathbf{i} - 4\mathbf{j}) = 12\mathbf{i} + 8\mathbf{j}$

M1: Adding forces to find resultant.
A1: Correct resultant force.

M1A1

2

(b) $4\mathbf{a} = 12\mathbf{i} + 8\mathbf{j}$ or $(\mathbf{a} =) \frac{12\mathbf{i} + 8\mathbf{j}}{4}$

*M1: Use of Newton's second law with 4**a** and their answer to part (a).*

M1

($\mathbf{a}$ =) 3**i** + 2**j** **AG**

A1: Correct acceleration from correct equation.

A1

2

(c) (i) 40**i** + 32**j** = **v** + (3**i** + 2**j**) × 20

40**i** + 32**j** = **v** + 60**i** + 40**j**

B1: Seeing 60i + 40j or (3i + 2j) × 20

B1

*M1: Use of constant acceleration equation with t = 20 and **a** = 3**i** ± 2**j**.*

M1

v = (40**i** + 32**j**) – (60**i** + 40**j**)

AG

= –20**i** – 8**j**

A1: Correct velocity from correct working, with one of the intermediate lines of working (or equivalent) shown.

*Note: Candidates may use **u** instead of **v** in their working.*

Example:

Starting with

***v** = 40**i** + 32**j** + (3**i** + 2**j**) × 20*

Scores B1M1A0.

A1

3

Note on Verification Method:

***v** = (–20**i** – 8**j**) + (3**i** + 2**j**) × 20 B1M1*

*= (–20 + 60)**i** + (–8 + 40)**j***

*= 40**i** + 32**j** A1*

Similarly, verification to confirm acceleration from the two velocities is acceptable.

(ii) (**v** =) (–20**i** – 8**j**) + (3**i** + 2**j**)

B1: Correct velocity vector.

*Note "**v** =" does not need to be seen.*

B1

1

(iii) (**v** =) (3t – 20)**i** + (2t – 8)**j**

*M1: Velocity vector seen split into components. Condone omission of **i** and **j***

Note: This can be implied by later working, such as the second line of this solution.

M1

$(3t - 20)^2 + (2t - 8)^2 = 8^2$

dM1: Equation based on speed of 8.

dM1

A1: Correct unsimplified equation.

A1

$13t^2 - 152t + 400 = 0$

A1: Simplified quadratic equation

A1

$t = \dfrac{152 \pm \sqrt{152^2 - 4 \times 13 \times 400}}{2 \times 13}$

dM1: Solving quadratic equation, to obtain two solutions.

dM1

$t = 4$ or $t = 7.69$

A1: Both correct solutions. Accept

AWRT 7.7 or 7.6 or $\dfrac{100}{13}$.

Note: Using calculator to solve quadratic is acceptable.

A1

6

[14]

Q4.

(a) $5 = \dfrac{1}{2} \times 9.8t^2$

M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

A1: Correct equation

A1

$t = \sqrt{\dfrac{5}{4.9}} = 1.01\text{ s}$ **AG**

A1: Correct time from correct working.

Must see square root or t^2 = 1.02 OE
Note: If g = 9.81 is used for the first time deduct one mark. Should still get 1.01 seconds.

A1

3

(b) $15 = V \times \sqrt{\frac{5}{4.9}}$

M1: Using distance = speed × time OE

M1

$V = 15\sqrt{\frac{4.9}{5}} = 14.8$

A1: Correct speed.
Accept AWRT 14.8 or 14.9.
Note: If g = 9.81 is used for the first time deduct one mark. Should get 14.9 m s^{-1} from g = 9.81.

A1

2

(c) $v_v = \pm 9.8 \times \sqrt{\frac{5}{4.9}} (= \pm 9.899)$

M1: Calculating vertical component of velocity.
A1: Correct value. Accept 9.9 or similar

M1A1

or

$v_v = \sqrt{2 \times 9.8 \times 5} = 9.899$

$v = \sqrt{9.899^2 + 14.8^2} = 17.8 \text{ m s}^{-1}$

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

dM1

A1: Correct speed.
Accept AWRT 17.8 or AWRT 17.9.

A1F

Note: If g = 9.81 is used for the first time deduct one mark. Should get 17.9 m s^{-1} from g = 9.81

4

(d) $\tan \alpha = \frac{9.899}{14.8}$ or $\frac{14.8}{9.899}$

$\alpha = 34°$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

3

$\sin \alpha = \frac{9.899}{17.8}$ or $\frac{14.8}{17.8}$

$\alpha = 34°$

(M1)(A1F)(A1F)

$\cos \alpha = \frac{14.8}{17.8}$ or $\frac{9.899}{17.8}$

$\alpha = 34°$

(M1)(A1F)(A1F)

(e) Particle

B1: Particle assumption

B1

Experiences no air resistance **or** no wind **or** only gravity **or** no other forces acting **or** no spin.

B1: Other assumption.
Ignore any other assumptions.

B1

2

[14]

Q5.

(a) $12g - T = 12a$

M1: Three term equation of motion, with $12g$ (or 117.6), $12a$ (not $12ga$) and T.
A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T.
A1: Correct equation

M1A1

$4g = 20a$

$a\left(= \dfrac{4g}{20}\right) = 1.96 \text{ m s}^{-2}$ AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is – 1.96 don't award final A1 mark.

Special Case:
Whole String Method $4g = 20a$ = and

$a = \dfrac{4g}{20} = 1.96$ *OE M1A1A1*

5

(b) $T = 8g + 8 \times 1.96 = 94.1$ N

M1: Use of three term equation of motion to find T, with $a = 1.96$.
A1: Correct tension.
Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v, with $a = 1.96$ and $u = 0$.
A1: Correct v.
Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v, with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity *from (c)(i).*

A1F

v = 9.68 m s^{-1}

A1F: Correct v. FT initial velocity from (c)(i).
For example 11.8 from 7.84.

A1F

3

Use of g = 9.81

(a) 1.962	*M1A1M1A1A0*
(b) 94.2	*M1A1*
(c) (ii) 9.69	*M1A1A1*

[12]

Q6.

(a) $\mathbf{v} = 9\mathbf{i} + 7\mathbf{j}$

B1: Correct initial velocity.

B1

1

(b) $\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$

*M1: Writing **v** in the form **v** = **u** + **a**t*

M1

$= (9\mathbf{i} + 7\mathbf{j}) + (-0.01\mathbf{i} - 0.03\mathbf{j})t$

$\mathbf{a} = -0.01\mathbf{i} - 0.03\mathbf{j}$

A1: Correct acceleration.

A1

2

(c) $9 - 0.01t = -(7 - 0.03t)$

*M1: Equation involving both **i** and **j** components of their velocity.*
A1: Correct equation, from their acceleration.

M1A1

$$t = \frac{16}{0.04} = 400$$

A1: Correct time, from their acceleration.

A1

3

[6]

Q7.

(a) 30 seconds

B1: Correct statement of time.

B1

1

(b) $s_1 = \frac{1}{2} \times 40 \times 20 = 400\text{ m}$

M1: A method for calculating the first distance. Must see 40 and $\frac{1}{2}$ *.*

M1

A1: Correct distance.

A1

OR

$s_1 = \frac{1}{2} \times (20 + 0) \times 40 = 400\text{ m}$

(M1)(A1)

OR

$a = -\frac{20}{40} = -\frac{1}{2}$

$0^2 = 20^2 + 2\left(-\frac{1}{2}\right)s$

$s = 20^2 = 400\text{ m}$

Note on third method: Must see $-\frac{1}{2}$ *or* $-\frac{20}{40}$ *plus attempt to find distance for M1.*

(M1)(A1)

2

(c) $s_2 = \frac{1}{2} \times 50 \times 20 = 500\text{m}$

M1: Method for finding the second distance and calculating the total distance.

M1

OR

$s_2 = \frac{1}{2} \times (0 + 20) \times 50 = 500\text{ m}$

(M1)

OR

$a = \frac{20}{50} = \frac{2}{5}$

$20^2 = 0^2 + 2\left(\frac{2}{5}\right)s$

$s = 20^2 \times \frac{5}{4} = 500 \text{ m}$

Note on third method: Must see $\frac{2}{5}$ *or* $\frac{20}{50}$ *plus attempt to find distance.*

(M1)

Total = 400 + 500 = 900 m

A1F: Correct total distance. Award the follow through mark for correct addition of 500 and their answer to (b).

A1F

2

(d) $v_{AVERAGE} = \frac{900}{120} = 7.5 \text{ ms}^{-1}$

M1: Their total distance divided by 120

M1

A1F: Correct average speed based on their answer to (c).

A1F

2

(e) 120 × 20 – 900 = 1500 m

M1: Multiplication of 20 and 120 to find distance.
Note: Award M1 if 2400 seen in this part.
A1F: Correct difference based on their answer to (c) provided final answer is positive.

M1A1F

2

[9]

Q8.

(a) $12g - T = 12a$

M1: Three term equation of motion, with

$12g$ (or 117.6), $12a$ (not $12ga$) and T.
A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with
$8g$ (or 78.4), $8a$ (not $8ga$) and T.
A1: Correct equation

M1A1

$4g = 20a$

$a\left(=\frac{4g}{20}\right) = 1.96 \text{ m s}^{-2}$ AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is – 1.96 don't award final A1 mark.

Special Case:
Whole String Method $4g = 20a$ and

$a = \frac{4g}{20} = 1.96$

OE M1A1A1

5

(b) $T = 8g + 8 \times 1.96 = 94.1$ N

M1: Use of three term equation of motion to find T, with $a = 1.96$.
A1: Correct tension.
Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v, with $a = 1.96$ and $u = 0$.
A1: Correct v.
Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v, with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$v = 9.68 \text{ m s}^{-1}$

A1F: Correct v. FT initial velocity from (c)(i). For example 11.8 from 7.84.

A1F

3

(iii) $4 = \frac{1}{2}(-3.92 + 9.68)t$

M1: Use of $s = \frac{1}{2}(u + v)t$

M1

A1: Correct values.

A1

A1: Correct signs.

A1

$t = 1.39$

dM1: Solving for t.

dM1

A1: Correct t.

A1

OR

$-4 = 3.92t - 4.9t^2$

$4.9t^2 - 3.92t - 4 = 0$

$t = \frac{3.92 \pm \sqrt{3.92^2 - 4 \times 4.9 \times (-4)}}{2 \times 4.9}$

M1: Forming a quadratic with candidates u from (c)(i) or v from (c)(ii) with 4.9 or 9.8.

(M1)

A1: Correct terms in quadratic.

(A1)

A1: Correct signs in quadratic.

(A1)

t = 1.39 or t = –0.588

dM1: Solving quadratic (do not penalise for negative discriminant).

(dM1)

t = 1.39

A1: Correct root seen (other root does not need to be seen).

(A1)

OR

$t_{up} + t_{down}$ = 0.4 + 0.4 + 0.588

M1: Finding total time from two or three times.

(M1)

A1: 0.4 or 0.8 seen.

(A1)

dM1: Finding second or third time for downward motion.

(dM1)

A1: Obtaining 0.588 or 0.988.

(A1)

= 1.39 (to 3SF)

A1: 1.39. Accept 1.38.

(A1)

OR

9.68 = –3.92 + 9.8t

M1: Use of $v = u + at$

(M1)

A1: Correct values.

(A1)

A1: Correct signs.

(A1)

$$t = \frac{13.6}{9.8} = 1.39$$

dM1: Solving for t

(dM1)

A1: Correct t

(A1)

5

Use of g = 9.81
(a) 1.962 M1A1M1A1A0
(b) 94.2 M1A1
(c) (ii) 9.69 M1A1A1
(c) (iii) 1.39 M1A1A1dM1A1

[17]

Q9.

(a) 10**a** = 9**i** + 12**j**

M1: Application of Newton's second Law with m = 10 in vector form.

M1

a = (0.9**i** + 1.2**j**) m s^{-2}

A1: Correct acceleration.
If acceleration incorrect follow their value through for the rest of this question.

A1

2

(b) (i) **r**(5) = (2.2**i** + 1**j**) × 5 + $\frac{1}{2}$ (0.9**i** + 1.2**j**) × 5^2

*M1: Use of constant acceleration to find position vector at t = 5, with **u** ≠ 0 **i** + 0 **j**.*

M1

= 22.25**i** + 20**j**

*A1F: Correct position vector, for candidate's acceleration which must be a vector. Allow 22.3**i** + 20**j**.*

A1F

$d = \sqrt{22.25^2 + 20^2}$ = 29.9 metres

dM1: Calculation of distance from position vector. Must see + sign.

dM1

*A1F: Correct distance, for their acceleration. Accept 30 from 22.3**i** + 20**j**.*

A1F

4

(ii) **v** = (2.2**i** + 1**j**) + (0.9**i** + 1.2**j**)t

M1: Use of constant acceleration equation to find an expression for **v**, *with* ***u*** *≠ 0* ***i*** *+ 0* ***j***

M1

A1F: Correct ***v*** *for their acceleration.*

A1F

2

(iii) $\mathbf{v} = (2.2 + 0.9t)\mathbf{i} + (1 + 1.2t)\mathbf{j}$

$2.2 + 0.9t = 1 + 1.2t$

$1.2 = 0.3t$

$t = 4$

M1: Equation involving both ***i*** *and* ***j*** *components of their velocity. Could have incorrect signs, for example*
$2.2 + 0.9t = -(1 + 1.2t)$.

M1

A1F: Correct equation.

A1F

A1F: Correct time, for their acceleration.

A1F

3

[11]

Q10.

(a) $14.7\sin\alpha - 9.8t (= 0)$

M1: Equation for vertical velocity being zero at highest point. Must have sin α with ± 9.8.
A1: Correct equation.

M1A1

$t = \dfrac{14.7\sin\alpha}{9.8} = \dfrac{3\sin\alpha}{2}$ **AG**

A1: Correct result from correct working.

A1

OR

$14.7\sin\alpha T - 4.9T^2 (= 0)$

$T = \dfrac{14.7\sin\alpha}{4.9} = 3\sin\alpha$

$t = \dfrac{3\sin\alpha}{2}$

All marks awarded for last line, from correct working.

(M1)(A1)(A1) 3

(b) (i) $7 = 14.7\sin\alpha\left(\dfrac{3\sin\alpha}{2}\right) - 4.9\left(\dfrac{3\sin\alpha}{2}\right)^2$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with ±g or equivalent.

M1

A1: Correct expression.

A1

$7 = 11.025\sin^2\alpha$

dM1: Simplified expression with $\sin^2\alpha$*.*

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

$\alpha = \sin^{-1}\left(\sqrt{\dfrac{7}{11.025}}\right) = 52.8°$

A1: Correct angle.
Accept 52.7°, 52.9°.

A1

OR

$0^2 = (14.7\sin\alpha)^2 + 2\times(-9.8)\times 7$

(M1)(A1)

$\sin^2\alpha = \dfrac{2\times 9.8\times 7}{14.7^2}$

(dM1)

$\alpha = 52.8°$

(dM1)(A1) 5

(ii) OA = 14.7 cos 52.8° × 3 sin 52.8°

B1: Use of 3sin α with their α.
M1: Finding horizontal displacement.

including 14.7 cos α with 3 sin α or $\frac{3\sin\alpha}{2}$ *.*

B1M1

OA = 21.2 m
A1: Correct distance. Accept 21.3 m.

A1

3

(c) Ball is a particle/No spin.
B1: Particle assumption.

B1

No air resistance/No wind/Constant acceleration of 9.8/Only force is weight.
B1: Air resistance assumption.

B1

2

[13]

Use of g = 9.81:

(a)	*M1A1A0*
(b) (i) 52.8° or 52.9°	*M1A1dM1dM1A1*
(b) (ii) 21.2	*B1M1A1*

Q11.

(a) $11g - T = 11a$
M1: Equation of motion for A, containing T, $11g$ or 107.8 and $11a$.

M1

A1: Correct equation

A1

$T - 9g = 9a$
M1: Equation of motion for B containing T, $9g$ or 88.2 and $9a$.

M1

A1: Correct equation

A1

$2g = 20a$

AG

a = 0.98 m s^{-2}

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$a = 2g / 20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(b) (i) $v = 0 + 0.98 \times 0.5 = 0.49$ m s^{-1}

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225$ m

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$0.49^2 = 0^2 + 2 \times 0.98s$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v.

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$d = 2 \times 0.1225$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

= 0.245 m

A1: Correct final distance. Allow 0.244 or 0.246. (Use of 0.5 × 0.49 = 0.245 scores zero unless justified)

A1

4

[11]

Q12.

(a) $0^2 = (30 \sin 35°)^2 + 2 \times (-9.8)s$

M1: Equation to fine the max height, with = 0

M1

A1: Correct equation

A1

M1: Solving for the height

M1

$$s = \frac{(30 \sin 35°)^2}{2 \times 9.8} = 15.1 \text{ m}$$

A1: Correct height

A1

4

(b) $28^2 = (30 \cos 35°)^2 + v_y^2$

M1: Equation to find vertical component

M1

A1: Correct equation

A1

$$v_y = \sqrt{28^2 - (30 \cos 35°)^2}$$

M1: Solving equation

M1

$= 13.4198 = 13.4 \text{ m s}^{-1}$ **AG**

A1: Correct speed from correct working.

A1

4

(c) $\tan\theta = \dfrac{13.4}{30\cos 35°}$

M1: Expression to find the angle.

M1

$\theta = 28.6°$

A1: Correct angle.

A1

2

[10]

Q13.

(a) $t = 0$, $t = 30$, $t = 50$ seconds

B1: Any one correct time

B1

B1: The other two correct times
Deduct one mark for each extra time if more than three times are given.
(eg 0, 15, 30, 50 scores B1 B0)
(eg 0, 15, 30, 40, 50 scores B0 B0)
Condone 49 or 48 instead of 50

B1

2

(b) $s_1 = \dfrac{1}{2} \times 30 \times 5 = 75 \text{ m}$ **AG**

M1: Finding distance by calculation of area. (Must see use of 0.5 or ½)

M1

A1: Correct answer from correct working.
(If candidates use two constant acceleration equations, both must be seen for the M1 mark.)

A1

2

(c) $s_2 = \dfrac{1}{2} \times 4 \times 20 = 40 \text{ m}$

M1: Finding distance using area of the second triangle.

M1

A1: Correct distance (ignore any negative igns). (If candidates use two constant acceleration equations, both must be seen for the M1 mark.)
Accept 38/36 from use of 49/48 instead of 0

A1

$s = 75 + 40 = 115$ m

M1: Addition of the 75 metres and their distance. (75 – 40 = 35 OE scores M0)

M1

A1F: Correct result using their value for second area.
eg Accept 113/111 from use of 49/48 instead of 50

A1F

4

(d) $s = 75 - 40 = 35$ m

M1: Difference between 75 and their value for the second distance.
(Allow their istance – 75)
(75 – (– 40) = 115 OE scores M0)

M1

A1F: Correct result using their value for second area.
(eg 40 – 75 = – 35 M1A0)
eg Accept 37/39 from use of 49/48 instead of 50

A1F

2

[10]

Q14.

(a) Peg is smooth

B1: Correct assumption

B1

1

(b) String is light

B1: First correct assumption

B1

String is inextensible or inelastic

B1: Second correct assumption

B1

Tension is the same throughout the string

Note: Ignore any additional assumptions.

2

(c) $11g - T = 11a$

M1: Equation of motion for A, containing, $11g$ or 107.8 and $11a$.

M1

A1: Correct equation

A1

$T - 9g = 9a$

M1: Equation of motion for B containing, $9g$ or 88.2 and $9a$.

M1

A1: Correct equation

A1

$2g = 20a$

AG

$a = 0.98 \text{ m s}^{-2}$

A1: Correct acceleration from correct orking.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and
$a = 2g/20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(d) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225$ m

M1: Finding distance travelled by each particle with u = 0, a = 0.98 and t = 0.5.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$0.49^2 = 0^2 + 2 \times 0.98s$

M1: Finding distance travelled by each particle with u = 0, a = 0.98 and their v

(M1)

$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$d = 2 \times 0.1225$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

= 0.245 m

A1: Correct final distance. Allow 0.244 or 0.246. (Use of 0.5 × 0.49 = 0.245 scores zero unless justified)

A1

4

If candidates calculate the distance first award marks as above (see (d) (i)) or:
M1: Use of constant acceleration equation to find v with u = 0, a = 0.98 and s = 0.1225.
A1: Correct v
Note: If parts (i) and (ii) are not separated or clearly labelled still award marks for both parts if justified.

[14]

Q15.

(a) Resultant = (6**i** – 3**j**) + (3**i** + 15**j**)
= 9**i** + 12**j**

M1: Summing the two vectors

M1

A1: Correct resultant

A1

2

(b) Magnitude $= \sqrt{9^2 + 12^2}$
= 15 N

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) 1.5m = 9 or 2m = 12

m = 6 kg m = 6 kg

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$$\frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6$$

*or 6**i** + 6**j** etc without a correct previous statement gives M0A0*

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with $\boldsymbol{u} = 0\boldsymbol{i} + 0\boldsymbol{j}$

M1

A1: Correct position vector.

A1

2

(ii) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$

$d = \sqrt{(3)^2 + (4)^2}$

M1: Finding the position vector when $t = 2$
($\mathbf{r} = (1.5\mathbf{i} + 2\mathbf{j}) \times 2 = 3\mathbf{i} + 4\mathbf{j}$ scores M0 unless it is clear how the 2 was obtained, possibly by a correct formula in (d) (i))

M1

$= \sqrt{25} = 5$

A1: Correct distance

A1

2

[10]

Q16.

(a) Resultant Force = 300 × 2.2
= 660 N AG

B1: Correct value from correct multiplication.

B1

1

(b) $P - 400 = 660$
$P = 1060$

M1: Three term equation of motion

M1

A1: Correct value for P

A1

2

(c) $23 = 12 + 2.2t$

M1: Use of a constant acceleration equation to find t.

M1

A1: Correct equation

A1

$$t = \frac{23 - 12}{2.2} = 5 \text{ s}$$

A1: Correct time

A1

3

[6]

Q17.

(a) $13^2 = 0^2 + 2 \times 1.3s$

M1: Use of a constant acceleration equation to find distance.

M1

A1: Correct equation

A1

$$s = \frac{13^2}{2.6} = 65 \text{ m}$$

A1: Correct distance

A1

3

(b) (i) $3900 - 800 - P = 2000 \times 1.3$

M1: Four term equation of motion for car and trailer.

M1

A1: Correct equation

A1

$P = 3900 - 800 - 2600 = 500$ N

A1: Correct value for P

A1

3

(ii) $T - 500 = 600 \times 1.3$

M1: Three term equation of motion for trailer.

M1

A1: Correct equation

A1F

$T = 500 + 780 = 1280$ N

A1: Correct tension

A1F

3

[9]

Q18.

(a) $16 = \frac{1}{2}(u + 4.2) \times 5$

M1: Using a constant acceleration equation to find u with v = 4.2 and $a \neq 9.8$. Could be derived from a velocity–time graph.
A1: Correct equation

M1A1

$32 = 5u + 21$

$5u = 11$

$u = \frac{11}{5} = 2.2 \text{ m s}^{-1}$

A1: Correct value for u

Eg $s = \frac{1}{2}(u + v)t$ *followed by*
16 = (u + 4.2) × 5 or similar scores M1A0

A1

OR
First solution from (b) to find acceleration followed by any constant acceleration equation to find u: eg.

(M1)

$4.2 = u + 0.4 \times 5$

(A1)

$u = 2.2$

(A1)

3

(b) $4.2 = 2.2 + 5a$

M1: Using a constant acceleration equation to find a with $u \neq 0$.
A1F: Correct equation. Follow through

for their incorrect u.

M1A1F

$5a = 2$

$a = \frac{2}{5} = 0.4\ \text{m s}^{-2}$

A1F: Correct value for a, which must be > 0.
Follow through for their incorrect u.
(If acceleration found correctly in part (a) and simply quoted as answer to (b) give full marks).

A1F

OR

$16 = 2.2 \times 5 + \frac{1}{2} \times a \times 5^2$

(M1)(A1F)

$16 = 11 + 12.5a$

$a = \frac{5}{12.5} = 0.4\ \text{m s}^{-2}$

(A1F)

OR

$16 = 4.2 \times 5 - \frac{1}{2} \times a \times 5^2$

(M1)(A1F)

$16 = 21 - 12.5a$

$a = \frac{5}{12.5} = 0.4\ \text{m s}^{-2}$

(A1F)

OR

$4.2^2 = 2.2^2 + 2a \times 16$

(M1)(A1F)

$a = \frac{17.64 - 4.84}{32} = 0.4\ \text{m s}^{-2}$

(A1F)

3

[6]

Q19.

(a) $R = 14 \times 9.8 = (137.2)$

M1: Finding the normal reaction. Accept 14g.

M1

$F = 0.25 \times 137.2$ **OR** $F = 0.25 \times 14 \times 9.8$

M1: Use of $F = \mu R$

M1

$F = 34.3$ N

A1: Correct friction
Use of g = 9.81 gives
R = 137.3 and F = 34.3 so in this case do not penalise use of g = 9.81.

A1

3

(b) $6g - T = 6a$

M1: Equation of motion for the particle, containing T, 6g or 58.8 and 6a.
A1: Correct equation with correct signs.

M1 A1

$T - 34.3 = 14a$

M1: Equation of motion for the block, containing T, 34.3 or their F and 14a.
A1: Correct equation with correct signs.

M1 A1

$6g - 34.3 = 20a$

$a = \dfrac{6g - 34.3}{20} = 1.225 \text{ m s}^{-2}$ AG

A1: Correct acceleration from correct working.
If – 1.225 is obtained from consistent working award 4 marks and if changed to +1.225 with an explanation, award full marks.
Special Case:
Whole string method

$6g - 34.3 = 20a$
$a = 1.225$ OE
award M1A1A1
Use of $g = 9.81$ gives
$a = 1.228$ penalise use of
$g = 9.81$ by deducting 1 mark, but don't penalise again on the same script.

(c) $T - 34.3 = 14 \times 1.225$

M1: Use of either of candidates equations of motion to find tension, with $a = \pm 1.225$ and their F (Method 1).

M1

$T = 17.15 + 34.3 = 51.5$ N

A1: Correct tension
Accept 51.45 or 51.4. Don't penalise use of $g = 9.81$ if already done in part (b).

A1

OR

$6g - T = 6 \times 1.225$

(M1)

$T = 6 \times 9.8 - 6 \times 1.225 = 51.5$

(A1)

2

(d) $v^2 = 0^2 + 2 \times 1.225 \times 0.8$

M1: Use of constant acceleration equation to find speed with $u = 0$.
A1: Correct equation

M1A1

$v = \sqrt{1.96} = 1.4 \text{ ms}^{-1}$

A1: Correct speed AWRT 1.4

A1

OR

$0.8 = \frac{1}{2} \times 1.225t^2$

In method 2, no marks awarded for just finding t.

$t = (1.1428)$

(M1)

$v = 1.225 \times 1.1428$

(A1)

$= 1.40$

(A1)

3

(e) $v^2 = 1.4^2 + 2 \times 9.8 \times 0.5$

M1: Use of constant acceleration equation to find speed with u = 1.4 or their answer to part (d), a = ±9.8 and s = 0.5.
A1F: Correct equation.
Follow through their answer to part (d).

M1A1F

$v = 3.43 \text{ m s}^{-1}$

A1F: Correct speed.
Don't penalise use of g = 9.81 if already done earlier in question.
In method 2, no marks awarded for just finding t.

A1F

OR

$0.5 = 1.4t + 4.9t^2$

$t = 0.2071$

$v = 1.4 + 9.8 \times 0.2071$

(M1)(A1F)

$= 3.43 \text{ m s}^{-1}$

(A1F)

3

[16]

Q20.

(a) **v** = (–2**i** + 2**j**) + (0.25**i** + 0.3**j**) × 20

*M1: Finding velocity using **v** = **u** + **a**t*

M1

A1: Correct expression

A1

$\mathbf{v} = 3\mathbf{i} + 8\mathbf{j}$

A1: Correct velocity in simplest form

A1

3

(b) $-2 + 0.25t = 0$

M1: One component equal to zero (either ***i*** *or* ***j*** *component).*
A1: Correct equation

M1A1

$t = 8$ s

A1: Correct time

A1

$\mathbf{v} = (2 + 0.3 \times 8)\mathbf{j} = 4.4\mathbf{j}$

A1: Correct velocity

A1

4

(c) $\mathbf{r} = (-2\mathbf{i} + 2\mathbf{j}) \times 20 + \frac{1}{2}(0.25\mathbf{i} + 0.3\mathbf{j}) \times 20^2 + (9\mathbf{i} + 7\mathbf{j})$

M1: Finding position vector using a constant acceleration equation with or without the initial position with $t = 20$.
A1: Correct expression for position vector including initial position.

M1A1

OR

$\mathbf{r} = \frac{1}{2}((-2\mathbf{i} + 2\mathbf{j}) + (3\mathbf{i} + 8\mathbf{j})) \times 20 + (9\mathbf{i} + 7\mathbf{j})$

$\mathbf{r} = 19\mathbf{i} + 107\mathbf{j}$

A1: Correct position vector in simplest form.

A1

3

(d) $\mathbf{v}_{AVERAGE} = \dfrac{(19\mathbf{i} + 107\mathbf{j}) - (9\mathbf{i} + 7\mathbf{j})}{20}$

M1: Finding average velocity based on change of position. Subtraction of initial *position must be seen or implied. Division by 8 scores M0*

M1

$$= \frac{10\mathbf{i} + 100\mathbf{j}}{20}$$
$$= 0.5\mathbf{i} + 5\mathbf{j}$$

A1F: Correct average velocity. Follow through incorrect answers from part (c).

Allow $\frac{\mathbf{u} + \mathbf{v}}{2}$

A1F

2

[12]

Q21.

(a) R = 15 × 9.8 = 147 N

F = 0.2 × 147 = 29.4 N

Finding normal reaction.

M1

Use of $F = \mu R$

M1

Correct friction force.

A1

3

(b) $5g - T = 5a$

Three term equation of motion for 5 kg particle

M1

Correct equation

A1

$T - 29.4 = 15a$ **AG**

Three term equation of motion for 15 kg block

M1

$49 - 29.4 = 20a$

Correct equation

A1

a = 0.98 m s^{-2}

Correct final answer from correct working.

A1

5

(c) $1.4^2 = 0^2 + 2 \times 0.98s$

$s = \frac{1.4^2}{1.96} = 1\,\text{m}$

Use of a constant acceleration equation with u = 0

M1

Correct equation

A1

Correct distance

A1

3

[11]

Q22.

(a) $8 = \frac{1}{2}a \times 5^2$

Use of constant acceleration equation with u = 0 to find a.

M1

$a = \frac{2 \times 8}{25} = 0.64\,\text{m s}^{-2}$ **AG**

Correct answer from correct working, showing evidence of solving for a. Allow verification / substitution.

A1

2

(b) $T - 70 \times 9.8 = 70 \times 0.64$

Three term equation of motion for crate.

M1

Correct equation

A1

T = 730.8 = 731 N to 3 sf

Correct tension

A1

3

(c) $v = \frac{8}{5} = 1.6 \text{ m s}^{-1}$

Correct average speed.

Accept $\frac{8}{5}$

Allow $\frac{3.2 + 0}{2} = 1.6 \text{ m s}^{-1}$

B1

1

[6]

Q23.

(a) (i) $0.2a = -0.2 \times 9.8 \sin 20°$

AG

$a = -9.8 \sin 20° = -3.35 \text{ m s}^{-2}$

Two term equation of motion with weight resolved

M1

Correct equation

A1

Correct acceleration from correct working

A1

SC No negative sign but otherwise correct award M1A1A0

Allow $a = -g \sin 20°$

3

(ii) $0 = 4^2 + 2 \times (-3.35)s$

Use of constant acceleration equation with v = 0 and u = 4

M1

Correct equation

A1

$s = \frac{16}{6.7} = 2.39 \text{ m}$

Correct distance

A1

3

(iii) The puck slides back down the slope as the puck is at rest and the resultant force is now acting down the slope / no friction / smooth slope.

Slides back down

B1

Acceptable explanation

E1

2

(b) (i) $R = 0.2 \times 9.8 \cos 20°$ **AG**

Finding normal reaction by resolving. Must see a trig term.

M1

$F = 0.5 \times 0.2 \times 9.8 \cos 20°$

Use of $F = \mu R$

M1

$= 0.921$ N

Correct friction from correct working.

A1

3

(ii) $0.2a = -0.921 - 0.2 \times 9.8 \sin 20°$

Three term equation of motion with the weight resolved

M1

Correct equation

A1

$a = -7.96$ m s^{-2}

Correct acceleration (with or without the minus sign, applied to both A1 marks)

A1

3

(iii) The puck stays at rest because the friction has a maximum of 0.921 and the component of the weight down the slope is less (0.670)

Stays at rest

B1

Acceptable explanation

dE1

2

[16]

Q24.

(a) $4\mathbf{i} = 5\mathbf{j} + 40\mathbf{a}$ **AG**

$\mathbf{a} = \dfrac{4\mathbf{i} - 5\mathbf{j}}{40} = 0.1\mathbf{i} - 0.125\mathbf{j}$

Forming a vector equation based on constant acceleration

M1

Correct equation

A1

Solving for ***a***

dM1

Correct ***a*** *from correct working*

A1

For $\dfrac{4\mathbf{i} - 5\mathbf{j}}{40}$ *on its own give M0*

Allow verification

4

(b) $\mathbf{r} = 5\mathbf{j} \times 40 + \frac{1}{2}(0.1\mathbf{i} - 0.125\mathbf{j}) \times 40^2$

$= 80\mathbf{i} + 100\mathbf{j}$

Finding position vector

M1

Correct expression

A1

Correct simplified result

A1

3

(c) (i) $\mathbf{v} = 5\mathbf{j} + (0.1\mathbf{i} - 0.125\mathbf{j})t$

$= 0.1t\mathbf{i} + (5 - 0.125t)\mathbf{j}$

Expression for ***v***

M1

*Correct expression for **v** seen or implied*

A1

$5 - 0.125t = -0.1t$

$5 = 0.025t$

Equating components, with or without a minus sign

dM1

Correct equation

A1

$t = \frac{5}{0.025} = 200$

Correct time.

A1

5

(ii) $\mathbf{v} = 0.1 \times 200\mathbf{i} + (5 - 0.125 \times 200)\,\mathbf{j}$

$= 20\mathbf{i} - 20\mathbf{j}$

Finding velocity using their time

M1

Correct velocity for their time

A1F

2

Note: Consistent use of u = 4i or 5i or a = 0.1i + 0.125j award method marks only.

[14]

Q25.

(a) $s = \frac{1}{2}(3 + 10) \times 3$

Finding distance by summing 3 areas or using formula for the area of a trapezium

M1

Correct equation/3 correct expressions for the areas

A1

= 19.5 m

Correct total distance

A1

3

(b) $a = \frac{3}{4} = 0.75\,\text{m s}^{-2}$

Correct acceleration as a decimal or as a fraction

B1

1

(c) $T - 400g = 400 \times 0.75$

Three term equation of motion containing T, $400g$ and 400×0.75 or equivalent

M1

Correct equation

A1F

$T = 3920 + 300 = 4220$ N

Correct tension

A1F

Only ft from $a = \frac{4}{3}$

(ft 4453 N or 4450 N from $a = \frac{4}{3}$ *scores M1A1A1)*

3

[7]

Q26.

Use of column vectors is acceptable throughout this question.

(a) $\mathbf{v} = 20\mathbf{i} + (-0.4\mathbf{i} + 0.5\mathbf{j})t$

Use of constant acceleration equation to find expression for ***v***

M1

Any correct expression.

A1

2

(b) $\mathbf{v} = (20 - 0.4t)\mathbf{i} + 0.5t\mathbf{j}$

Simplifying ***v****. (May be implied.)*
(Missing brackets may be condoned if followed by correct working.)

M1

$20 - 0.4t = 0$

*Putting **i** component equal to zero*

m1

$t = \frac{20}{0.4} = 50$ seconds

Correct time
Candidates who are able to see the correct time without supporting working gain full marks.

Condone $\frac{20\mathbf{i}}{0.4\mathbf{i}} = 50$

A1

3

(c) $\mathbf{r} = 20\mathbf{i} \times t + \frac{1}{2}(-0.4\mathbf{i} + 0.5\mathbf{j}) \times t^2$

*Use of constant acceleration equation to find expression for **r***

M1

Any correct expression

A1

2

(d) (i) $\mathbf{r} = 20\mathbf{i} \times 100 + \frac{1}{2}(-0.4\mathbf{i} + 0.5\mathbf{j}) \times 100^2$

*Substituting t = 100 into their expression for **r** (dependent on M1 in part (c))*

m1

$= 2000\mathbf{i} - 2000\mathbf{i} + 2500\mathbf{j}$
$= 2500\mathbf{j}$

*Correct simplified position vector ie 2500**j***

A1

Therefore due north

Conclusion that helicopter is due north provided their position vector is of the form $k\mathbf{j}$, where $k > 0$
Note if integration is used there is no need to prove that the constant is zero.
Note marks for (d) (i) can be awarded if part (c) scores zero.

A1

3

(ii) $\mathbf{v} = (20 - 0.4 \times 100)\mathbf{i} + 0.5 \times 100\mathbf{j}$

*Substituting t = 100 into their expression for **v** (dependent on M1 in part (a)) or use of other constant acceleration equation and their position vector (dependent on M1 in part (c))*

m1

$= -20\mathbf{i} + 50\mathbf{j}$

Correct simplified velocity

A1

$$v = \sqrt{20^2 + 50^2} = 53.9$$

Correct speed (Accept $10\sqrt{29}$)

A1

Note marks for (d) (ii) can be awarded if part (a) scores zero.

3

[13]

Q27.

(a) $16^2 = 12^2 + 2a \times 20$

Use of CA equation(s) to find a

M1

Correct equation for a

A1

$$a = \frac{16^2 - 12^2}{40} = 2.8 \text{ m s}^{-1}$$

Correct acceleration

A1

3

(b) $16 = 12 + 2.8t$

Use of CA equation(s) to find t

M1

Correct equation for t

A1

$t = \frac{16 - 12}{2.8} = 1.43$ s

Correct t

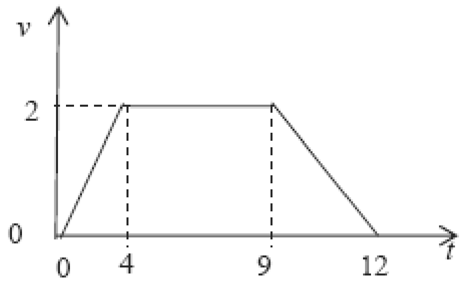
A1

3

[6]

Q28.

(a)



Starts and finishes at rest

B1

Correct shape

B1

Correct values on t-axis

B1

Correct values on v-axis

B1

Condone omission of the origin

4

(b) $s = \frac{1}{2}(5 + 12) \times 2$

or $s = \frac{1}{2} \times 2 \times 4 + 5 \times 2 + \frac{1}{2} \times 2 \times 3 = 17$

Use of the area under the graph (or equivalent) to find s

M1

= 17

Correct distance
SC *When 21 used instead of 12 allow full marks for s = 26*

A1

2

(c) max $a = \frac{2}{4} = 0.5$

Maximum acceleration

B1

$300 \times 0.5 = T - 300 \times 9.8$

Three term equation of motion using their a

M1

Correct equation using a = 0.5

A1

$T = 2940 + 150 = 3090$

Correct tension

A1

4

[10]

Q29.

(a) The string is light and inextensible or inelastic or taut

First assumption

B1

Second assumption

B1

2

(b) $6 = 0 + 4a$

Finding a using a CA equation

M1

$a = \frac{6}{4} = 1.5$

Correct a from correct working

A1

2

(c) $7 \times 9.8 - T = 7 \times 1.5$

Three term equation of motion with F for the 7 kg particle. Correct equation

M1A1

$T = 68.6 - 10.5 = 58.1$

Correct tension

A1

3

(d) $58.1 - F = 13 \times 1.5$

Three term equation of motion with F for the 13 kg particle. Correct equation

M1A1

$F = 58.1 - 19.5 = 38.6$

Correct F

A1

$R = 13.98 = 127.4$

Correct R

B1

$38.6 = \mu \times 127.4$

Use of $F = \mu R$

dM1

$$\mu = \frac{38.6}{127.4} = 0.303$$

Correct coefficient of friction

A1

6

[13]

Q30.

(a) $0^2 = (50 \sin 40°)^2 + 2 \times (-9.8)h$

Equation for h with v = 0 and a component of velocity. Correct equation

M1A1

$$h = \frac{(50 \sin 40°)^2}{2 \times 9.8} = 52.7$$

Solving for h

dM1

Correct h

A1

Alt
$0 = 50 \sin 40° - 9.8t$

Equation for t with v = 0 and a component of velocity

(M1)

$t = \dfrac{50 \sin 40°}{9.8} = 3.280$

Correct t

(A1)

$h = 50 \sin 40° \times 3.280 - \dfrac{1}{2} \times 9.8 \times 3.280^2$

Expression for h with a component of velocity

(dM1)

= 52.7
ALLOW 52.6

Correct h

(A1)

4

(b) $6 = 50 \sin 40° t - 4.9t^2$

Forming a quadratic in t. Correct terms with any signs

M1A1

$0 = 4.9t^2 - 50 \sin 40° t + 6$

Correct equation

A1

$t = \dfrac{50 \sin 40° \pm \sqrt{(50 \sin 40°)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$

Solving quadratic

dM1

= 0.192 or 6.37
t = 6.37

Correct solution selected

A2

Alt

$46.7 = 4.9 t_1^2$

Finding two times

(M1)

Equation for time to go down

(dM1)

$t_1 = 3.087$

Correct time

(A1)

$t_2 = 3.280$

Time to go up

(A1)

$t = 3.087 + 3.280 = 6.37$

Correct total

(A2)

6

[10]

Q31.

(a) $75\mathbf{i} = (5\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2}\mathbf{a} \times 10^2$

Equation to find ***a*** *from* $\boldsymbol{r} = \boldsymbol{u}t + \frac{1}{2}\boldsymbol{a}t^2$

M1

Correct expression

A1

$\mathbf{a} = \dfrac{75\mathbf{i} - 50\mathbf{i} + 20\mathbf{j}}{50} = 0.5\mathbf{i} + 0.4\mathbf{j}$

AG *Correct* ***a*** *from correct working*

A1

3

(b) $\mathbf{r} = (5\mathbf{i} - 2\mathbf{j}) \times 8 + \frac{1}{2}(0.5\mathbf{i} + 0.4\mathbf{j}) \times 8^2$

Expression for ***r*** *using* $t = 8$ *with no extra terms*

M1

Correct expressions

A1

$= 56\mathbf{i} - 3.2\mathbf{j}$

Correct position vector

A1

3

(c) $\mathbf{v} = (5 + 0.5t)\mathbf{i} + (0.4t - 2)\mathbf{j}$

Expression for ***v***. *Correct expression*

M1A1

$0.4t - 2 = 0$

j *component equal to zero*

dM1

$t = \frac{2}{0.4} = 5$

Correct t

A1

$\mathbf{r} = (5\mathbf{i} - 2\mathbf{i}) \times 5 + \frac{1}{2}(0.5\mathbf{i} + 0.4\mathbf{j}) \times 5^2$

Expression for ***r*** *using t from* ***j*** *component equal to zero*

dM1

$= 31.25\mathbf{i} - 5\mathbf{j}$
$= 31.3\mathbf{i} - 5\mathbf{j}$

Correct position vector

A1

6

[12]

Q32.

(a) $320 = \frac{1}{2} \times a \times 80^2$

Use of constant acceleration equation with $u = 0$

M1

Correct equation

A1

$a = \frac{2 \times 320}{80^2} = 0.1 \text{ m s}^{-2}$

AG Correct acceleration from correct working

A1

3

(b) $v = 0 + 0.1 \times 80$

Use of constant acceleration equation with $u = 0$

M1

$= 8 \text{ m s}^{-1}$

Correct velocity

A1

2

(c) $L - 450 \times 9.8 = 450 \times 0.1$

Three term equation of motion

M1

Correct equation

A1

$L = 45 + 4410$
$= 4455$
$= 4500$ N (to 3sf)

AG Correct force

A1

3

(d) 4410 N

Correct force

B1

1

[9]

Q33.

(a) $3.45g - T = 3.45a$

Three term equation of motion for one particle

M1

Correct equation

A1

$T - 1.45g = 1.45a$

Three term equation of motion for other particle

M1

Correct equation

A1

$2g = 4.9a$

$a = \frac{2 \times 9.8}{4.9} = 4\text{ m s}^{-2}$

AG Correct acceleration from correct working

A1

5

(b) $T = 1.45 \times 4 + 1.45 \times 9.8$

Use of one equation from (a) to find T

M1

$= 20.01$

$= 20.0$ N (to 3 sf)

Correct T

A1

2

(c) $s = \frac{1}{2} = 0.5\text{ m}$

Use of $s = \frac{1}{2}$

M1

$v^2 = 2 \times 4 \times 0.5$

Use of constant acceleration equation with $u = 0$

M1

$v = 2\text{ m s}^{-1}$

Correct speed (no negative sign)

A1

3

[10]

Q34.

(a) $2.45 = \frac{1}{2} \times 9.8t^2$

Equation for time to ground

M1

Correct equation

A1

$t = \sqrt{\frac{2.45}{4.9}} = 0.707$ seconds (to 3 sf)

AG Correct time from correct working

A1

3

(b) $s = 20 \times 0.707 = 14.1$ m (to 3sf)

Using 20 × time from part (a)

M1

Correct distance

A1

2

(c) $v_y = 0.707 \times 9.8 = 6.929$

Finding vertical component of velocity

M1

Correct vertical component

A1

$\theta = \tan^{-1}\left(\frac{6.929}{20}\right) = 19.1°$

Using tan to find the angle

dM1

Correct angle

A1

4

[9]

Q35.

(a) $v = 0 + 1.5 \times 9.8$

Use of constant acceleration equation to find v

M1

$= 14.7 \text{ m s}^{-1}$

AG Correct v from correct working
$1.5 \times 9.8 = 14.7$ is not enough on its own

A1

2

(b) $h = \frac{1}{2} \times 9.8 \times 1.5^2$

Use of constant acceleration equation with
$a = 9.8$ to find h

M1

$= 11.0$ m (to 3 sf)

Correct h
Allow 11 m; ignore negative signs

A1

2

(c) $5^2 = 0^2 + 2 \times 9.8s$

Use of constant acceleration equation with
$u = 0$ to find s

M1

Correct equation

A1

$$s = \frac{25}{19.6} = 1.28 \text{ m(to 3 sf)}$$

Correct s
Accept 1.27

A1

OR

$$t = \frac{5}{9.8} = 0.510$$

$$s = \frac{1}{2}(0 + 5)\frac{5}{9.8} = 1.28 \text{ m}$$

OR

$$s = 0 + \frac{1}{2} \times 9.8 \times \left(\frac{5}{9.8}\right)^2 = 1.28\text{ m}$$

3

[7]